Commutative Algebra

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Michaelmas 2023*

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Lecture 1

In this course, a ring is a commutative unital ring R. One non-commutative exception is the ring $\mathrm{End}(M)$, where M is an abelian group. This is a ring with pointwise addition, and composition as multiplication.

Definition 0.1 (module)
An *R-module* M is an abelian group M with an fixed ring homomorphism $\rho : R \to \mathrm{End}(M)$. We will write $r \cdot m := \rho(r)(m)$.

Remark 0.2. By definition, this implies that $(r_1 + r_2) \cdot m = r_1 \cdot m + r_2 \cdot m$, $r \cdot (m_1 + m_2) = r \cdot m_1 + r \cdot m_2$ and $r_1 \cdot (r_2 \cdot m) = (r_1 r_2) \cdot m$.

Example 0.3 (Examples of modules)

- Let k be a field. Then a k-module is the same as a k-vector space.
- Every abelian group is a $\mathbb{Z}$-module in a unique way, since we must have that $\rho(1) = \mathrm{id}_M$. Therefore, abelian groups and $\mathbb{Z}$-modules are the same thing.
- Every ring R is (trivially) an R-module.
- More generally, $R^{\oplus \mathbb{N}}$ (direct sum) and $R^{\mathbb{N}}$ (direct product) are R-modules.

1 Chain conditions

Definition 1.1 (Noetherian, Artinian module)
An R-module M is *Noetherian* if one of the following (equivalent) conditions hold:

1. Every ascending chain of submodules $M_0 \subseteq M_1 \subseteq M_2 \subseteq \cdots$ stabilises. That is, it is eventually constant.
2. Every non-empty set Σ of submodules of M has a maximal element.

M is *Artinian* if we replace in the above: ascending with descending, maximal with minimal.

Lemma 1.2. An R-module M is Noetherian if and only if every submodule of M is finitely generated.

In particular, every Noetherian module is finitely generated. If $R = \mathbb{Z}[T_1, T_2, \ldots]$, with $M = R$ as an R-module. Then M is finitely generated. On the other hand, $M' = \langle T_1, T_2, T_3, \ldots \rangle$, is not finitely generated.

*Based on lectures by Oren Becker. Last updated October 6, 2023.

Definition 1.3 (Noetherian, Artinian ring)
A ring R is Noetherian (resp. Artinian) if it is Noetherian (resp. Artinian) as an R-module.

Example 1.4
1. $\mathbb{Z}$ is Noetherian (as it is a PID), but not Artinian (e.g. $\langle 2\rangle \supseteq \langle 4\rangle \supseteq \langle 8\rangle \supseteq \cdots$).
2. $\mathbb{Z}[1/2]/\mathbb{Z}$ is Artinian, but not Noetherian as a $\mathbb{Z}$-module.
3. A ring R is Artinian if and only if R is Noetherian and R has Krull dimension 0.

Definition 1.5 (Exact sequence)
A sequence
$$\cdots \longrightarrow M_{i-1} \xrightarrow{f_i} M_i \xrightarrow{f_{i+1}} M_{i+1} \longrightarrow \cdots$$
of R-modules and R-module homomorphisms is *exact* if $\operatorname{im}(f_i) = \ker(f_{i+1})$ for all i.

Definition 1.6 (Short exact sequence)
A *short exact sequence* (SES) is an exact sequence of the form
$$0 \longrightarrow N \xhookrightarrow{\iota} M \twoheadrightarrow L \longrightarrow 0$$
That is, we have an embedding $\iota : N \hookrightarrow M$, and an isomorphism $L \cong M/\iota(N)$.

Lemma 1.7. Let
$$0 \longrightarrow N \longrightarrow M \longrightarrow L \longrightarrow 0$$
be an SES of R-modules. Then M is Noetherian (resp. Artinian) if and only if N and L are Noetherian (resp. Artinian).

Proof. We may assume without loss of generality that N is a submodule of M. Let $P_1 \subseteq P_2 \subseteq \ldots$ be an increasing (resp. decreasing) sequence of submodules of M. In this case,
$$N \cap P_1 \subseteq N \cap P_2 \subseteq \cdots$$
is an increasing (resp. decreasing) sequence of submodules of N, hence eventually constant. Similarly,
$$\frac{N + P_1}{N} \subseteq \frac{N + P_2}{N} \subseteq \cdots$$
is an increasing (resp. decreasing) sequence of submodules of $L = M/N$, hence eventually constant. For large n, we will have
$$P_n \subseteq P_{n+1} \quad N \cap P_n = N \cap P_{n+1} \quad N + P_n = N + P_{n+1}$$
Hence $P_n = P_{n+1}$ for large enough n. □

Corollary 1.8. If $M_1, \ldots, M_n$ are Noetherian (resp. Artinian) R-modules, then $M_1 \oplus \cdots \oplus M_n$ is Noetherian (resp. Artinian).

Proof. By the lemma and induction. □

Recall a module homomorphism
$$\varphi : M_1 \oplus \cdots \oplus M_n \to L$$
is the same as a collection of module homomorphism $\varphi_i : M_i \to L$. This is also true for infinite direct sums (but not products!).

Proposition 1.9. For a Noetherian (resp. Artinian) ring R, every finitely generated R-module is Noetherian (resp. Artinian).

Proof. M is finitely generated if and only if there exists a surjection $R^n \twoheadrightarrow M$ for some $n \in \mathbb{N}$. The fact that R^n is Noetherian (resp. Artinian) implies that M is Noetherian (resp. Artinian), as quotients of Noetherian (resp. Artinian) modules are Noetherian (resp. Artinian). □

Definition 1.10 (algebra)

An *R-algebra* A is a ring A with a fixed ring homomorphism $\rho : R \to A$. We will write $r \cdot a := \rho(r)a$.

Remark 1.11. Every R-algebra is an R-module.

Example 1.12

The polynomial ring $k[T_1, \ldots, T_n]$ is a k-algebra. Do note however that it is a finitely generated by $T_1, \ldots, T_n$ as a k-algebra, but it is infinite dimensional as a k-vector space.

Definition 1.13 (algebra homomorphism)

$\varphi : A \to B$ is an *R-algebra homomorphism* if φ is a ring homomorphism and $\varphi(r \cdot 1_A) = r \cdot 1_B$.

Definition 1.14 (finitely generated algebra)

An R-algebra A is *finitely generated* if there exists a surjective R-algebra homomorphism $R[T_1, \ldots, T_n] \twoheadrightarrow A$ for some $n \in \mathbb{N}$.

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